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Cross-scale sensing of field-level crop residue cover: integrating field photos, airborne hyperspectral imaging, and satellite data

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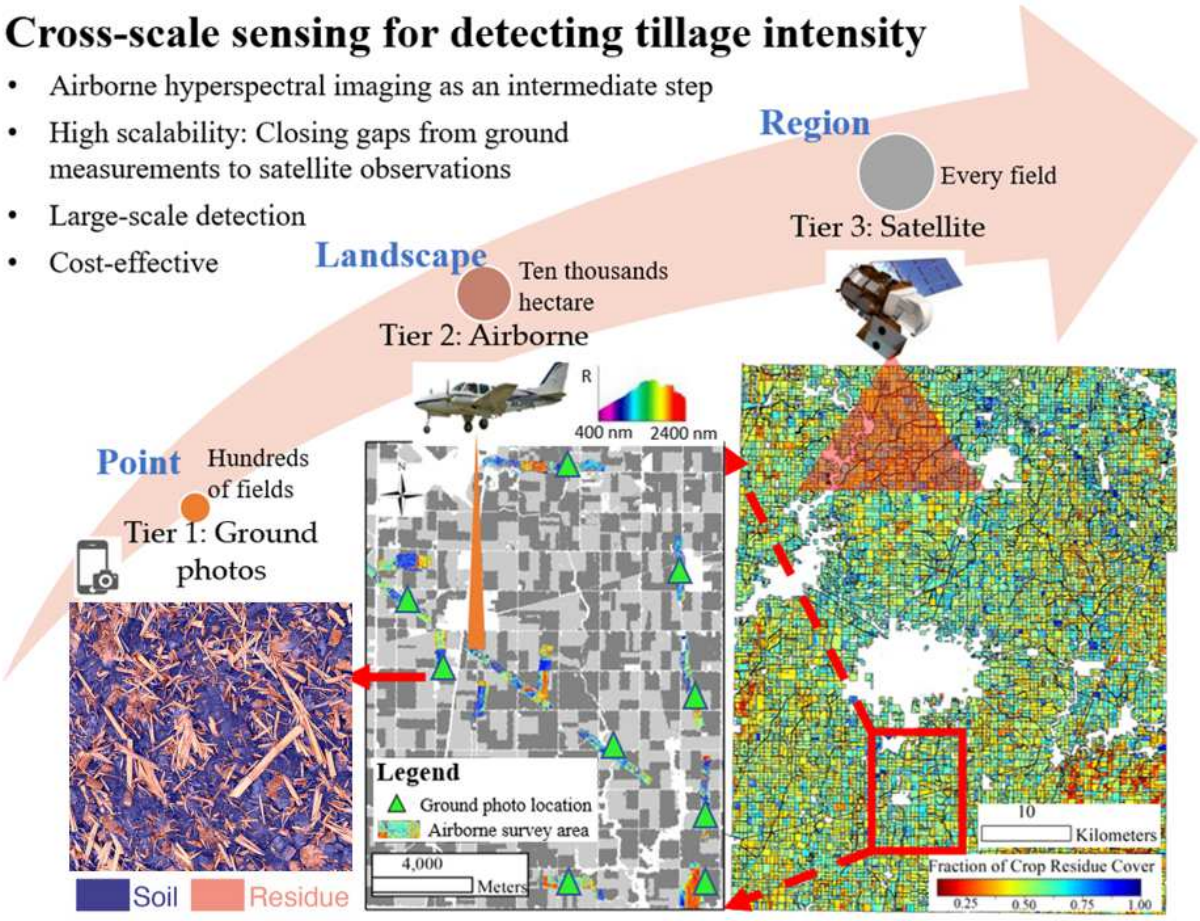
Abstract:

Conservation tillage practices can bring significant benefits to agricultural sustainability. Accurate spatial and temporal resolved information of field-scale crop residue cover, which reflects tillage intensity, is highly valuable for evaluating the outcomes of government conservation programs and voluntary ecosystem service markets, as well as facilitating agroecosystem modeling to quantify cropland biogeochemical processes. Remote sensing has the potential to cost-effectively detect crop residue cover, however, existing regional-scale studies were limited by insufficient ground truth data, scale mismatch between coarse satellite pixels and ground data, and the lack of key spectral data for detecting crop residues. Therefore, this study developed an innovative cross-sensing framework to integrate proximal sensing, airborne hyperspectral imaging, and satellite Earth Observation through deep learning to quantify field-level crop residue cover fractions at the regional scale. Specifically, we have collected intensive ground orthographic photos and conducted airborne hyperspectral surveys at corn and soybean fields of Champaign and nearby counties in Illinois, the heartland of the U.S. Corn Belt. Through semi-automatic labeling aided by ResNet-50 and superpixel image segmentation, we obtained 6,719 records of ground residue fractions. With these ground data, we developed the 1-dimensional convolution neural network (CNN) model using airborne hyperspectral reflectance, which has 0.5 m spatial resolution and 3–5 nm spectral resolution from 400 to 2400 nm, to predict residue fractions. By applying the CNN model to airborne pixels, we augmented "ground truth" data of crop residues and further combined them with Harmonized Landsat and Sentinel-2 (HLS) satellite data to quantify regional residue fractions at 30 m resolution. Results show that airborne hyperspectral imagery with CNN can accurately detect residue fractions (R^2 = 0.82, relative RMSE = 11.73%) to effectively generate quasi "ground truth" data to support satellite upscaling to all fields. With independent ground data for testing, we found that the ground-airborne-satellite integrative framework achieved better predictions in estimating crop residue cover (R^2 = 0.67, relative RMSE = 17.53%) than the conventional ground-satellite upscaling (R^2 = 0.22, relative RMSE = 32.09%). We also found that the shortwave infrared wavelengths, particularly 2100–2300 nm, are vital for predicting crop residue cover.

Sentinel-2 and Landsat-8 data have a comparable capability to track residue fractions due to similar shortwave infrared wavelengths. This study highlights the high accuracy of hyperspectral imaging to detect agroecosystem tillage management practices and the advantages of cross-scale sensing to cost-effectively integrate multi-source data to quantify field-level agroecosystem variables across scales.

Keywords: Crop residue, Tillage practices, Airborne hyperspectral imaging, Sentinel-2, Landsat, Convolutional Neural Networks, ResNet-50, Sustainable agriculture, Multi-scale remote sensing

Graphic abstract:



1. Introduction

Sustainable agriculture with maintaining or increasing high yields while minimizing negative environmental impacts is pivotal for feeding the world's growing population and for mitigating climate changes (IPCC, 2022). As one of the essential practices for sustainable agriculture, conservation tillage (e.g., reduced- and non-tillage) has been adopted to replace conventional (intensive) tillage around the world for the benefits of reducing soil erosion, increasing soil carbon and fertility, as well as, saving costs on labor, fuel, and machinery (Papendick and Parr, 1997; Deines et al., 2019). The adoption of conservation tillage can significantly alter cropland nutrient, carbon, water, and energy dynamics (Davin et al., 2014; Mehra et al., 2018) and has been considered an important management practice for soil carbon sequestration to mitigate climate change (Follett, 2001; Bai et al., 2019). As one of the most intensive cropping systems in the world, the U.S. Corn Belt contributes one-third of global corn and soybean production (USDA, 2017), but faces severe issues on agroecosystem sustainability (Thaler et al., 2021). Given the benefits of conservation tillage, reduced- and non-tillage practices have been promoted in the past decades by government conservation programs, such as USDA Conservation Reserve Program (CRP) and Environmental Quality Incentives Program (EQIP), and the recent emerging voluntary carbon markets (Oldfield et al., 2022). To assess the outcome of current policies and understand the factors influencing the adoption of conservation tillage, accurate spatially and temporally resolved information on cropland tillage practices is urgently needed. Furthermore, tillage intensity information is also essential for agroecosystem modeling to facilitate the accurate quantification of agroecosystem biogeochemical processes as well as cropland carbon outcomes (Maharjan et al., 2018; Guan et al., 2022). However, such information at the regional scale is still rare and only a few studies have identified regional tillage classes (e.g., two classes including high- and low-intensity tillage, Azzari et al., 2019).

The key physical variable that directly reflects the outcome of tillage practices is crop residue cover. The residue fraction refers to the percentage of non-photosynthetic senesced plant litter, such as corn stalks or

soybean stubble, accumulated on the ground. The fraction of crop residue cover is often used as the proxy of tillage intensity (Hively et al., 2018), and tillage types can be classified according to the ranges of crop residue cover (e.g., 0–15, 15–30, 30–60, >60%) after planting (CTIC, 2022). Similarly, USDA's National Agricultural Statistics Service (NASS) provides county-level three-class tillage types (i.e. conventional, reduced, and non-tillage acreage) based on residue cover (USDA, 2017).. To obtain accurate information on tillage practices, the field-level data on crop residue cover are, traditionally, collected through farmer self-reporting in governmental or commercial programs, or in-person field investigations such as the USDA line transect method (Morrison et al., 1993) and annual roadside surveys (CTIC, 2022). However, these survey data have limitations on data accuracy, data sharing, spatial and temporal coverage due to surveyors' personal judgment, farmers' concerns about data privacy, and the high cost of field surveys. Therefore, conventional field surveys are difficult to provide timely information on tillage intensity for every field across broad regions (Zheng et al., 2014) and, so far, USDA NASS only reports county-level tillage types once every five years. Given these limitations, there is an urgent need to develop a timely, cost-effective, and accurate approach for monitoring regional-scale tillage intensity.

Remote sensing provides opportunities to quantify crop residue cover to infer tillage intensity in a cost-effective manner. In the past decades, studies have been conducted with proximal (e.g., Nagler et al., 2003), airborne (e.g., Daughtry et al., 2006), and satellite (e.g., Azzari et al., 2019) sensing to quantify residue cover fractions or tillage intensity classes. In the principles of spectral signals in detecting crop residues, these studies found that crop residues and soils have similar spectral signatures in the visible and near-infrared regions, but residues show unique absorption features in the shortwave infrared region, particularly lignin and cellulose absorption bands in 2100 and 2300 nm (Asner and Lobell, 2000; Daughtry, 2001, Daughtry et al., 2004; Hively et al., 2018; Wang et al., 2022). As shortwave infrared wavelengths are highly valuable for crop residue detection, optical hyperspectral remote sensing with rich narrow shortwave-infrared wavelengths shows a high capability to detect residue cover. For example, airborne hyperspectral remote sensing with the full optical range of 400–2400 nm and high spatial resolution to be matched with

ground measurements can identify tillage classes with high accuracy of > 90% (Daughtry et al., 2006), which is much higher than the accuracy of multispectral sensing studies (e.g., 75–79%, Azzari et al., 2019). However, airborne hyperspectral data have limited spatial and temporal coverages, and are hard to be applied to quantify crop residue cover for every field across large regions.

Satellite remote sensing is more cost-effective to quantify regional crop residue cover for every field, compared to airborne sensing or field investigations. In the past years, studies have explored the utility of satellite data with shortwave infrared bands, such as Hyperion (Daughtry et al., 2006; Bannari et al., 2015), ASTER (Serbin et al., 2009), Landsat (Azzari et al., 2019), Sentinel-2 (Zhou et al., 2021), MODIS (Watts et al., 2011), and WorldView-3 (Hively et al., 2018), to quantify tillage intensity classes (e.g. conventional, reduced or non-tillage). Among these satellite data, the hyperspectral mission Hyperion achieved around 80% accuracy in distinguishing tillage classes for three counties in central Iowa (Daughtry et al., 2006). However, the limited spatial coverage, low frequency, and low signal-to-noise ratios significantly limited Hyperion for regional applications (Middleton et al., 2013). Multispectral satellite missions, e.g. ASTER and WorldView-3, with rich narrow bands in the shortwave infrared can also track tillage practices with high accuracy (Serbin et al., 2009; Hively et al., 2018). However, ASTER shortwave infrared sensors have failed since 2008 (Serbin et al., 2009) and the commercial satellite mission WorldView-3 has limited spatial and temporal coverage and a high cost. The coarse spatial resolution MODIS (i.e., 500m) has difficulty in detecting tillage practices for small fields. Among all satellite missions, Landsat and Sentinel-2 with two shortwave infrared wavelengths at 20–30m spatial resolution are often utilized for monitoring tillage intensity (Azzari et al., 2019; Yue et al., 2020; Gao et al., 2022). Particularly, multi-source satellite fusion or harmonized data, e.g. National Aeronautics and Space Administration (NASA) Harmonized Landsat Sentinel-2 (HLS, Claverie et al., 2018), can combine Landsat and Sentinel-2 to provide high temporal frequency data, which provide potential high-value data sources for quantifying regional tillage practices. However, existing studies have not fully realized the potential of these fusion data, and studies exploring their capability to quantify tillage practices are highly needed. Meanwhile, 30m multispectral satellite data

could have uncertainties to be matched with point measurements of crop residue cover in fields. How to seamlessly integrate remote sensing with ground data needs further exploration.

To overcome the above challenges in existing studies, we aim to develop an innovative cross-scale sensing framework to seamlessly integrate ground proximal sensing, airborne hyperspectral imaging, and satellite Earth Observation to accurately quantify regional residue cover. To derive regional information, the large-scale satellite data need to have a paired high volume of ground truth data for machine learning model development (Deines et al. 2021), particularly considering that soil sensing signals vary significantly in space and time. However, collecting such large amounts of ground truth data through conventional field investigation is highly labor- and cost-intensive. Thus, this cross-sensing framework innovatively deploys airborne hyperspectral imaging systems to collect sub-meter resolution surface reflectance with hundreds of wavelengths. As airborne hyperspectral remote sensing has a high accuracy to derive tillage intensity (Daughtry et al., 2006), we first upscale the ground photo-derived label data of crop residues to airborne hyperspectral pixels for croplands to obtain a large amount of quasi "ground truth". This airborne-hyperspectral-based quasi "ground truth" is then used to scale up the satellite algorithm to regional scale tillage mapping. As such, we not only augment ground truth data cost-effectively using airborne hyperspectral data, but also minimize the scale-mismatch issue by linking airborne data with satellite data for upscaling (instead of directly scaling in-situ ground data to satellite). This framework thus uses the multi-source harmonized satellite data, i.e., HLS reflectance, to upscale airborne-based quasi "ground truth" to every field in the region of interest. By doing so, we cost-effectively obtain the information of regional-scale residue cover.

Besides the innovations in integrating multi-scale sensing data, this cross-scale sensing framework also aims to leverage versatile computer vision and deep learning algorithms to apply to ground photos, airborne hyperspectral imagery, and satellite multispectral observations collectively to quantify crop residue cover.

Specifically, to accurately derive ground truth data of residue cover fractions, we leverage computer vision image segmentation algorithms to semi-automatically identify crop residues on the ground. As such, we can mitigate the uncertainties from manual labeling in field investigations. In remote sensing algorithms, conventional approaches to derive residue cover fractions primarily focused on multispectral indices operating with a few wavelengths, e.g. Normalized Difference Tillage Index (NDTI, Van Deventer, et al., 1997), Cellulose Absorption Index (CAI, Nagler et al., 2003; Daughtry, et al., 2004), Crop Residue Index Multiband (CRIM, Biard and Baret, 1997), Modified Soil Adjusted Corn Residue Index (MSACRI, Daunthery, et al., 1997), and Broadband crop residue Angle Index (BAI, Yue et al., 2020) as well as using random forest classifiers to incorporate spectral reflectance (e.g., Azzari et al., 2019, Watts et al., 2011). With the advances in artificial intelligence, new deep learning algorithms, such as convolutional neural networks (CNN) and multi-layer perceptron neural networks (MLP), have the high potential to make full use of spectral wavelengths (Wang et al., 2022) without making handcraft indices to predict residue cover fractions.

The general objective of this study was to develop an accurate, robust, and scalable approach to integrate proximal, airborne, and satellite sensing through deep learning to quantify regional-scale field-level fractions of crop residue cover to infer tillage intensity in croplands. The specific objectives were: (1) To develop and assess the proximal and airborne sensing to obtain a large volume of ground truth data for residue cover fractions. These sensing approaches include employing computer vision algorithms to semi-automatically identify residue cover fractions from close-range field photos, and leveraging airborne hyperspectral imaging with deep learning to augment field photo-derived residue fractions to obtain quasi-ground truth data at the regional scale; (2) To assess the performance of integrating multi-scale data from proximal, airborne, and satellite sensing for quantifying residue cover fractions, and analyze the added values of airborne hyperspectral sensing as the intermediate step for upscaling field data to satellites; and (3) To identify key spectral information for quantifying crop residue cover fraction and test the applicability of Sentinel-2 and Landsat-8 satellite data for quantifying regional tillage intensity.

2. Study region and data

2.1 Study region

This study focused on corn and soybean fields in Champaign and nearby counties, Illinois. As in the central region of the U.S. Corn Belt (Fig. 1a), this region is dominated by agricultural landscapes with more than 90% of fields planted with corn (Zea mays L.) and soybean (Glycine max [L.] Merr.). With a typical continental climate, corn and soybean fields in this region are typically planted in April–May and harvested in September–October. For conventional tillage practices, growers often conduct primary deep tillage to loosen soils within the depth of 20–30 cm in the fall right after the harvest, and a few farmers also conduct secondary tillage in spring after soil thawing to loosen soils at a depth of 5–8 cm to prepare the seedbed (Omonode et al., 2007). For reduced tillage practices, soil disturbance is much less, and residue cover fractions in the field are higher than in fields with conventional tillage, while fields with non-tillage practices are covered by high fractions of crop residues left after the harvest (Claassen et al., 2018). The major soil order and texture in the study region are Mollisols and silt loams, respectively. The soil organic carbon concentrations at surface layers (0–15 cm) are around 1–4% $g \cdot g^{-1}$ (Potash et al., 2022).

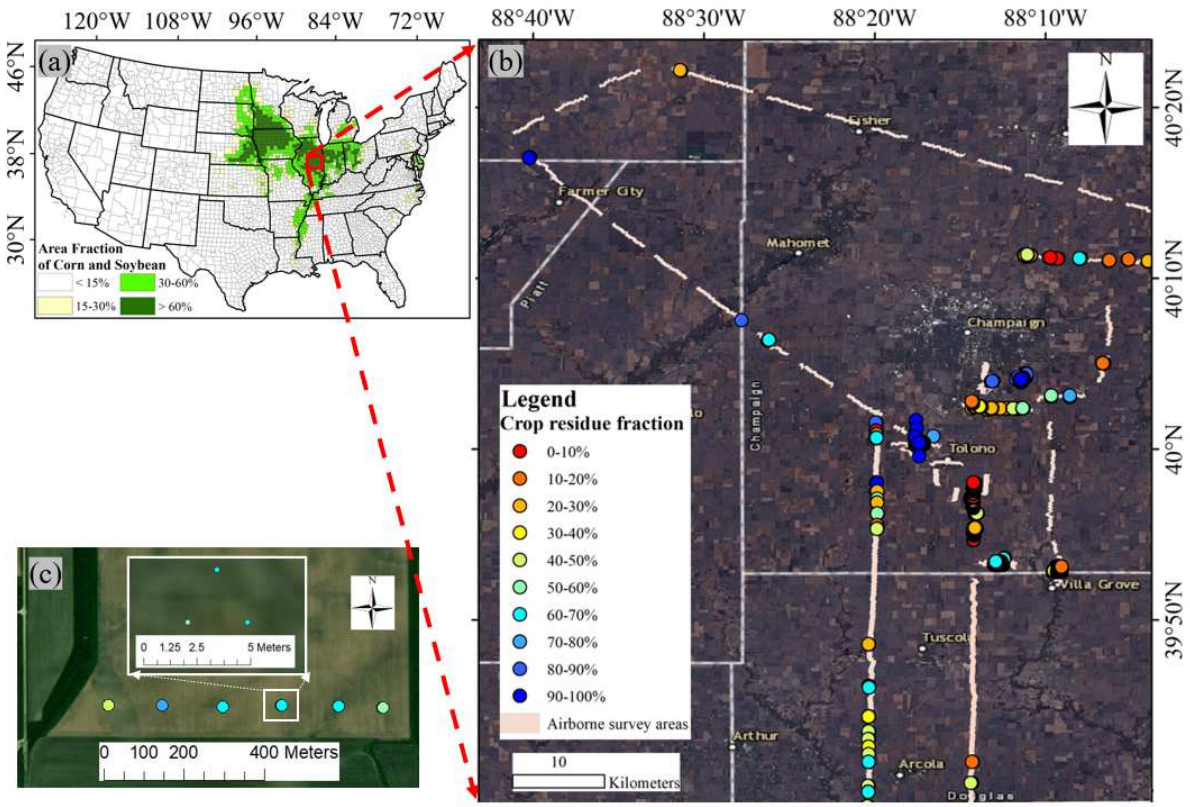


Fig. 1 Study region of Champaign County and nearby counties. (a) Corn and soybean planting area fractions across the U.S. The study region is in the center of the U.S. Corn Belt. (b) Study region map. The pink polygons refer to the collected airborne hyperspectral imagery. The circles are the locations of collected ground photos. The base map is the true color HLS imagery on December 26th, 2020. (c) Example of ground photo collection in one field.

2.2 Data

2.2.1 Field photos

We conducted field investigations to collect field photos in an orthographic view and then from these photos to derive highly accurate data on crop residue cover fractions. In the field measurement, we used cellphones to take photos with an approximately nadir position and a height roughly around 1.5 m above the ground. Each photo was geotagged with Global Navigation Satellite System (GNSS) location information, which

has an accuracy of 2–5 m in an open area environment for typical cellphones (Petovello et al., 2014; Van Diggelen et al., 2015). Fig. 1(b) shows the detailed locations for these ground photos. Specifically, we collected photos at 6–10 locations in each field as an example in Fig. 1(c). These locations have more than 30m distance between each other. In each location, we took 3–4 photos with a distance of 2–4m between each other. To avoid significant residue shadows covering soils, all photos were taken at a local time of 10am–4pm. To provide sufficient data on ground crop residue fractions for airborne hyperspectral sensing, we have collected a total of 6,719 ground photos. Among them, there are 3358 and 3361 records for the spring (within three days after April 21st, 2020) and winter (within three days after December 22nd, 2020) campaigns (Table 1), respectively.

2.2.2 Airborne hyperspectral data

Three airborne hyperspectral flight surveys were conducted by an aircraft (PA23-160, Piper Aircraft, US) on March 7th, April 21st, and December 22nd, 2020 (Table 1). The airborne hyperspectral system was equipped with HySpex VNIR-1800 (spectral range: 400–1000 nm) and SWIR-384 (spectral range: 950–2400 nm) imaging sensors (HySpex NEO, Norway) onboard to collect surface reflected radiance data over the study region (Fig. 1). VNIR-1800 and SWIR-384 sensors have the full wavelength at half maximum (spectral resolution) of 3.26 and 5.45 nm, respectively, and the same radiometric resolution of 16 bits. To acquire sensors' position and angular data in the flight, an Inertial Measurement Unit (IMU) and differential GNSS (Applanix, Trimble, US) were attached to the imaging sensors. The airborne system flew at an altitude of 1.5 km above the ground, which led to the acquired hyperspectral imagery with spatial resolution of 0.5 m. The airborne surveys were conducted with large solar elevation angles (see the local time in Table 1). In total, we approximately scanned 102 km^2 croplands across this region.

Table 1. Summary of airborne hyperspectral surveys in the study for quantifying crop residue cover.

Date	Local Time (US central time)	Weather	Relative humidity (%)	Air temperature (°C)	Start time solar elevation (°)	Noon time solar elevation (°)	End time solar elevation (°)	Date of satellite data for upscaling

Mar 7, 2020	10:31–13:52	Sunny	55.6	4.8	40.3	45.1	38.7	Mar 8, 2020
Apr 21, 2020	10:48–13:58	Sunny	30.9	10.2	58.9	62.1	50.7	Apr 20, 2020
Dec 22, 2020	10:10–14:33	Partially sunny	53.2	6.1	22.3	26.6	16.4	Dec 26, 2020

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236 **2.2.3 Satellite remote sensing**

237 For satellite remote sensing data, we used 30-m spatial resolution HLS data to scale up airborne estimates
238 of crop residues to every field across the study region. As two major public satellite missions by NASA and
239 European Space Agency (ESA), Landsat and Sentinel-2 provide freely available moderate-to-high spatial
240 resolution multispectral satellite data, and they are also the two most widely used in tillage classification.
241 The synergy of Landsat and Sentinel-2 data, such as HLS, creates unique opportunities for timely and
242 accurate detection of land surface changes. Specifically, in HLS data processing, Sentinel-2 MSI L1C and
243 Landsat-8 OLI L1 products were used as inputs to be processed through spatial coregistration, atmospheric
244 corrections, cloud masking, bandpass adjustment, and bi-directional reflectance normalization to generate
245 consistent nadir-adjusted surface reflectance for these two satellites (Claverie et al., 2018). HLS provided
246 per-pixel quality flags for cloud, shadow, snow, and water masks. However, the quality flag of HLS version
247 1.4 is problematic for Sentinel-2 (Bolton et al., 2020) and we used the Sentinel-2 quality flag from Google
248 Earth Engine to remove low-quality pixels in HLS Sentinel-2 (HLS S30). By excluding bands for detecting
249 atmospheric conditions, HLS S30 has 10 bands including blue (450–510 nm), green (530–590 nm), red
250 (640–670 nm), red edge 1 (690–710 nm), red edge 2 (730–750 nm), red edge 3 (770–790 nm), NIR Broad
251 (780–880 nm), NIR Narrow (850–880 nm), SWIR 1 (1570–1650 nm), and SWIR 2 (2110–2290 nm). HLS
252 Landsat (L30) has 6 bands including blue (450–510 nm), green (530–590 nm), red (640–670 nm), NIR
253 Narrow (850–880 nm), SWIR1 (1570–1650 nm), and SWIR2 (2110–2290 nm).

3. Methodology

The workflow of this study includes three data tiers, i.e., ground, airborne, and satellite data. To fill the spatial upscaling gaps and acquire massive highly accurate quasi "ground truth" of residue cover fractions, we conducted ground-airborne-satellite upscaling as Fig. 2. Meanwhile, we also conducted the direct upscaling from ground to satellite remote sensing. Specifically, from ground photos, we developed computer vision-aided algorithms, including Deep Residual Networks (ResNet-50, He et al., 2016) and superpixel image segmentation (Felzenszwalb & Huttenlocher, 2004), to semi-automatically label photos to derive crop residue cover (see Section 3.1). Furthermore, we compared two upscaling approaches from ground to satellite data to quantify regional crop residue cover. In the first approach, the ground-airborne-satellite integration used airborne hyperspectral imagery as the intermediate step to upscale ground crop residue cover to satellite data (see Section 3.3). This approach used ground data as labels and airborne hyperspectral data as features to develop CNN models (see Section 3.2). Then the developed models were applied to airborne imagery pixels to quantify crop residue cover fractions. The airborne derived fractions were then used as labels along with satellite data to develop MLP. By applying MLP to all satellite data, we can obtain the crop residue cover for the whole study region. In the second approach, the ground-satellite upscaling directly used the ground photo-derived crop residue cover as ground truth, and then upscaled to satellite data (see Section 3.4). Finally, we evaluated these two upscaling approaches to HLS using the independent ground measurements for model testing (see Section 3.6).

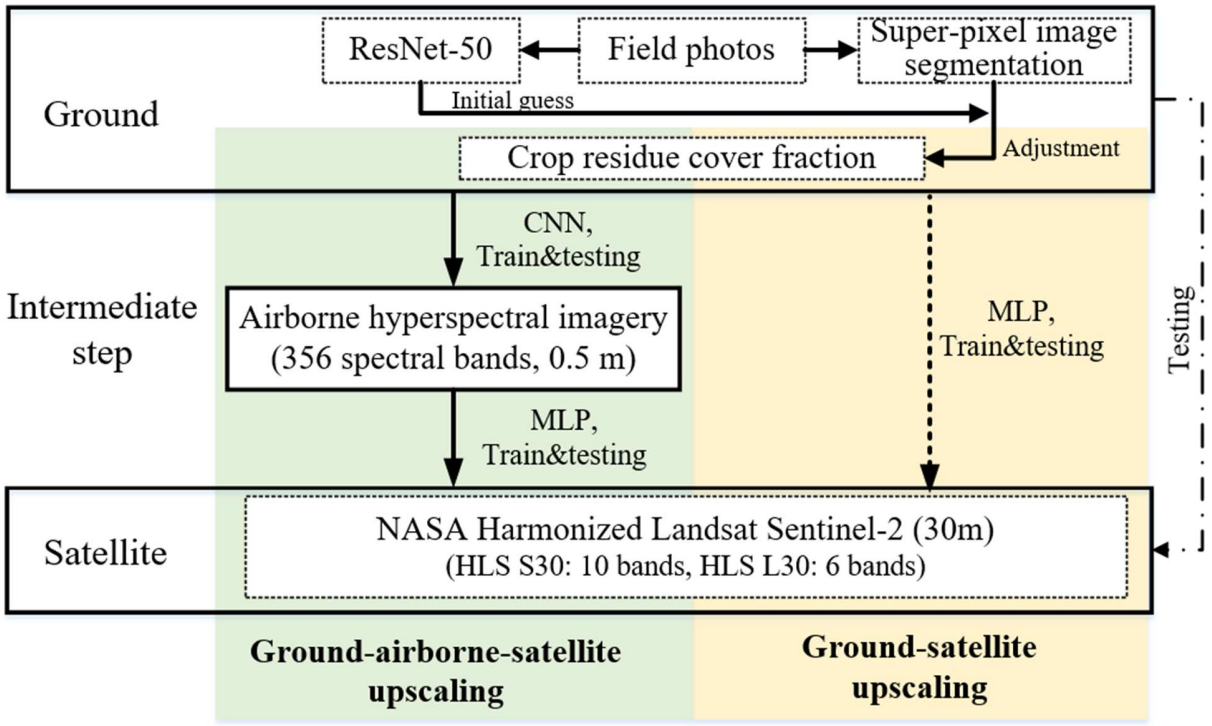


Fig. 2 Workflow of deriving crop residue cover fractions from ground, airborne, and satellite cross-scale sensing. ResNet-50 is a 50-layer-deep convolutional neural network (CNN) to provide the first guess of residue cover fraction. MLP stands for a multilayer perceptron model. S30 and L30 refer to 30-m Sentinel-2 and Landsat data, respectively.

3.1 Field photo processing

To obtain highly accurate ground crop residue cover, we developed a computer vision-aided semi-automatic labeling method (Fig. 3) to facilitate the labeling process. In the first step, we manually labeled 1317 photos to obtain the residue cover fractions. Using these residue fraction data as labels, we fine-tuned ResNet-50 (He et al., 2016), which was pre-trained on ImageNet (Deng et al., 2009), to predict crop residue fractions from photos. To quantify model uncertainty, we use Monte Carlo Dropout (Gal and Zoubin, 2016) to obtain a prediction interval for each image. To label new photos, we applied ResNet-50 to obtain the first guess of crop residue fraction. Traditionally, ResNet-50 was designed for image classification, and in this study, we modified ResNet-50 from a classifier to a regressor. Specifically, we still used the same backbone structure as in ResNet-50, but we changed the final classification layers, which were used to map the latent

representation to the prediction scores of a number of different classes, to output only one quantity — the predicted tillage residue. We finetuned both the ResNet-50 backbone pre-trained on ImageNet and the modified regressor on new ground tillage photos with Mean Squared Error (MSE) loss. Meanwhile, we also employed a superpixel segmentation algorithm, i.e., Felzenszwalb's method (Felzenszwalb & Huttenlocher, 2004), to partition photo pixels into "superpixels" — units that only cover a single object (but a single object can still consist of multiple superpixels). Then, we converted the RGB channel values to the greyscale of each superpixel. In the following step of manual photo labeling, the labeling person only needs to check and tune the residue fractions according to the first-guess residue fraction generated by ResNet-50 and image gray-scale histogram distribution (see the graphic interface in Fig. S1). As such, the labeling person selected the threshold to separate the average values of background and foreground superpixels. The final residue mask was eventually generated by selecting superpixels that were on the appropriate side of the threshold. Compared to direct labeling through simple thresholding, this approach can reduce noises and biases to improve labeling accuracy by incorporating ResNet-50 initial prediction and superpixel segmentation. Furthermore, this approach speeded up labeling processes and is much more efficient than the conventional labeling methods, e.g., manually marking every foreground pixel (or, equivalently, drawing the foreground boundary). By using this semi-automatic approach, we obtained 6,719 records of crop residue fractions from all collected ground photos.

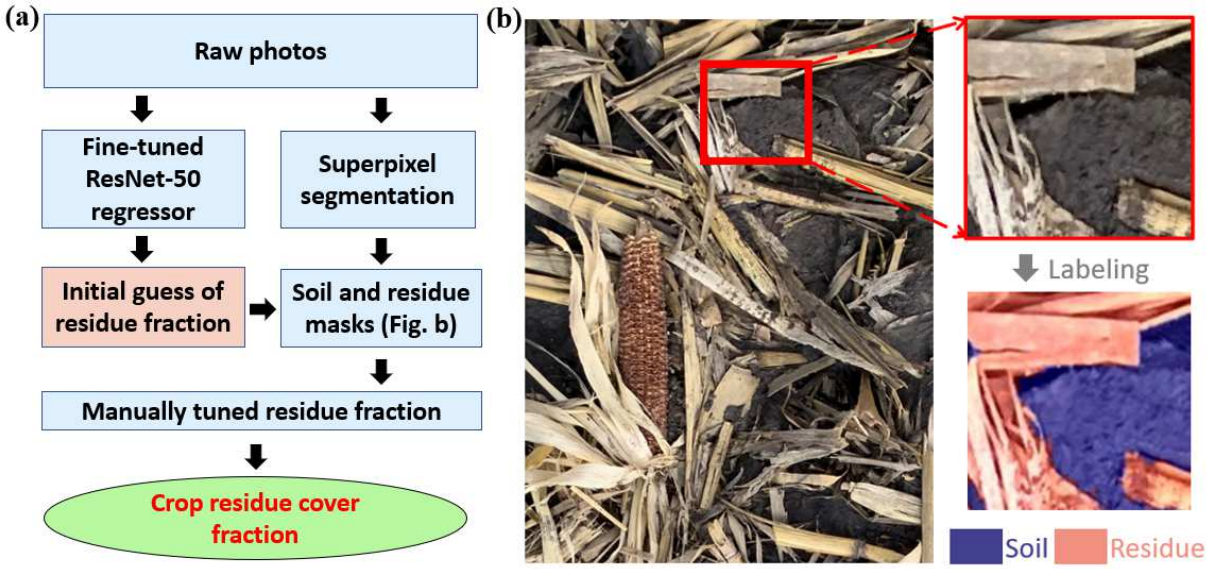


Fig. 3 (a) Procedures of processing field-collected photos to derive ground measurements of crop residue cover. (b) Example of deriving crop residue cover from field photos using ResNet-50 initial guess and superpixel segmentation.

3.2 Airborne data processing

To obtain high-quality surface reflectance, we followed Wang et al. (2021) procedures on geometric, radiometric, and atmospheric corrections to process airborne raw imagery. Specifically, we first converted airborne sensor raw imagery (digital numbers) to at-sensor radiance data using sensor radiometric calibration coefficients from the manufactory. Then, two popularly used airborne data processing software PARGE and ATCOR (Richter and Schläpfer, 2016) were used to process geometric and atmospheric corrections, respectively. In geometric corrections, PARGE used the flight recorded GNSS and IMU data, pre-calibrated sensor boresight coefficients, and surface digital elevation model (DEM) to conduct imagery orthorectification. After that, the atmospheric corrections of orthorectified airborne imagery were processed through ATCOR, which used the MODTRAN look-up table approach (Berk et al., 2005). In this atmospheric correction process, we selected the ATCOR mid-latitude, winter, and rural aerosol models and

removed the airborne data SMILE, keystones, and atmospheric attenuation effects. To correct the bidirectional reflectance distribution function (BRDF) effects induced by variable solar elevation angles and sensor view angles, we performed BRDF correction for airborne hyperspectral imagery using the open-source package HyTool (https://github.com/EnSpec/hytools, Liu et al., 2021). With these steps of data processing, we obtained high-quality surface reflectance with 356 spectral bands from 400 to 2400 nm, 3–5 nm spectral resolution and 0.5 m spatial resolution.

3.3 Ground-airborne-satellite upscaling

In the ground-airborne-satellite upscaling (Fig. 2), we first identified the paired data sets for airborne hyperspectral reflectance (0.5 m spatial resolution and 356 spectral bands) and the field photo-based residue cover fractions (from section 3.1). We used airborne hyperspectral data as feature inputs and field crop residue fractions as labels to develop a 1-dimensional CNN model. To account for the uncertainties of field photo GNSS data (2–5m), we have tested aggregated airborne spectral data from different zone sizes (change from 0.5m to 5.5m with an interval of 0.5m) in airborne pixels to identify the best match between airborne and field measurements. Specifically, we developed 11 CNN models with feature inputs of spectral data from aggregated airborne pixels (0.5–5.5m) and identified the aggregated sizes with the best model testing performance. Then, by applying this best airborne deep learning model to all airborne pixels, we can obtain massive quasi "ground truth" of crop residue fractions for croplands. After that, we will obtain the paired airborne estimates with satellite multispectral remote sensing (HLS S30, 30 m spatial resolution) to further develop the satellite-based MLP deep learning model. Finally, by applying the satellite MLP model to all cropland satellite pixels, we can get the estimates of regional-scale crop residue fractions. By applying this upscaling framework, we seamlessly integrate ground data, airborne hyperspectral sensing, and satellite multispectral Earth Observation to quantify crop residue fractions for every field across the targeted region.

3.4 Direct upscaling of ground measurements to satellite

As a comparison to the ground-airborne-satellite upscaling framework, we also conducted the direct upscaling from ground measurements to satellite multispectral observations (HLS S30, 30 m spatial resolution). In this process (Fig. 2), we first identified the geolocation-paired ground truth data with satellite remote sensing and then developed an MLP model to predict crop residue fractions from satellite spectral data. We conducted a thorough comparison with the ground-airborne-satellite upscaling.

3.5 Machine learning algorithms

We used two machine learning algorithms including 1-dimensional CNN and MLP to develop models to predict crop residue fractions from remote sensing data. Specifically, hyperspectral data have 356 wavelengths and we applied CNN to make full use of hyperspectral data to predict crop residue fractions. Meanwhile, for multispectral bands in satellite remote sensing data, we applied MLP to predict crop residue fractions. The details of these two algorithms are mentioned below.

MLP used generic nonlinear function approximation by building a model of the data-generating processes for the network to generalize and predict target variables (Hecht-Nielsen, 1989). In this study, MLP consisted of one input layer, one output layer, and at least one hidden layer. This study used a back-propagation neural network regression model to deal with highly nonlinear relationships between feature inputs and label outputs. The MLP model parameters including the optimal number of hidden layers and neurons were optimized through parameter screening to achieve the smallest RMSE. CNN has similar features of multiple fully connected layers as MLP, but adds extra convolutional layers before fully connected layers. Through convolution operation, CNN can account for connections among feature data, e.g., linkages across hyperspectral wavelengths, other than MLP.

To analyze the feature importance for the CNN and MLP deep learning models, we utilized the permutation approach. The permutation feature importance was determined by the decrease of a model evaluation score

(Root Mean Squares Errors, RMSE) when a single feature input (each spectral wavelength) was randomly shuffled (Altmann et al., 2010). This permutation feature importance has been widely applied for evaluating the importance of feature inputs in deep neural network models (e.g., Wang et al., 2022).

3.6 Model implementation and validation

In all machine learning model development of this study, we split the whole dataset into training (75% of all data) and testing (25% of all data). Notably, the model accuracy can be inflated with the testing and training data from the same fields (Zheng et al., 2012; Azzari et al., 2019). To avoid such cases, we consider field sources in the data splitting of ground measurements, which means that ground measurements from the same field did not occur in both training and testing datasets. In the process of deep learning model training testing, we utilized weight decay to mitigate potential overfitting. Concretely, weight decay penalizes model parameters with excessively large L2 norm (Loshchilov and Hutter, 2018). We also plotted figures of training vs testing loss curves to demonstrate that we did not encounter significant overfitting issues to hinder the generalization of the model. In model evaluation, we used statistics including coefficient of determination (R^2), RMSE, relative RMSE (rRMSE), bias, and relative bias. rRMSE was the ratio between RMSE and data range. The better model performance corresponds to the higher values of R^2 and the lower values of RMSE, rRMSE, and bias, or vice versa.

There are two benefits of airborne hyperspectral imagery to quantify tillage practices: the high spatial resolution of airborne imagery to be benchmarked with ground photo measurements, and rich high spectral information to support CNN deep learning predict crop residue cover. To understand the separated benefits from high spatial and rich spectral information for quantifying crop residue cover, we also used multiple spectral wavelengths to calculate NDTI (Eq. 1) and CAI (Eq. 2) from airborne hyperspectral data. Both NDTI and CAI are two most popular indices used in corn and soybean systems to quantify crop residue cover (Nagler et al., 2003; Daughtry, et al., 2004; Serbin et al., 2009; Hively et al., 2018; Hively et al., 2021). Furthermore, we also followed these NDTI and CAI studies to conduct Pearson correlation and linear regression analysis with the ground derived crop residue cover. In addition, we compared R^2 between

spectral indices and ground residue cover with the CNN deep learning approach, which utilized all spectral information in hyperspectral data. This comparison provides us insights into distinguishing contributions from high spectral and high spatial resolutions in high-performance airborne hyperspectral imagery for quantifying crop residue cover.

$$NDTI = \frac{R_{2200} - R_{1600}}{R_{2200} + R_{1600}} \quad \text{Eq. (1)}$$

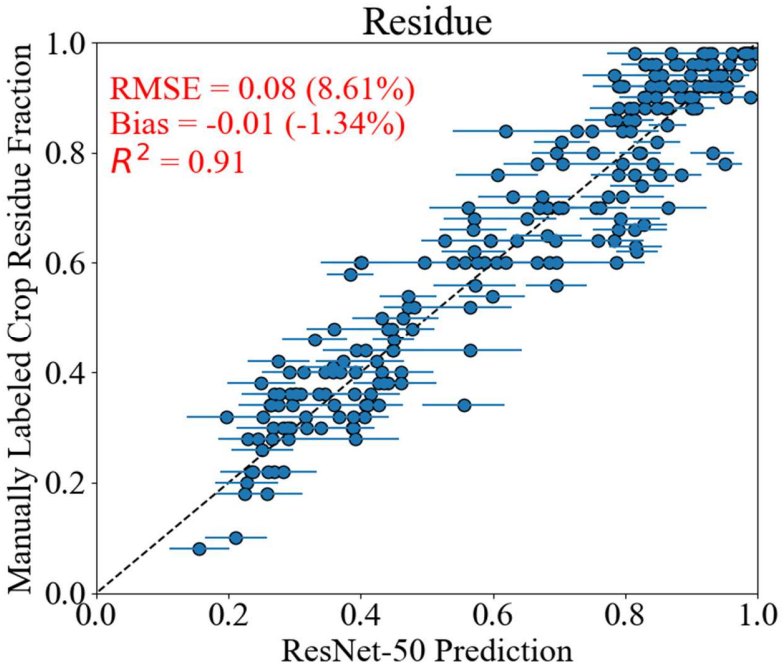
$$CAI = 0.5(R_{2000} + R_{2200}) - R_{2100} \quad \text{Eq. (2)}$$

Where R refers to surface reflectance at a specific wavelength. For example, R_{2200} is reflectance at the wavelength of 2200nm.

4. Results

4.1 Crop residue fractions derived from ground field photos

In the first step of semi-automatic labeling, ResNet-50 can effectively give high accuracy of estimates of crop residue fractions (Fig. 4). Through further superpixel image segmentation, grey-scale histogram distribution and manual checking, we obtained 6,719 records of ground photo-based crop residue fractions across the study region (Fig. 1). Fig. 5 (a) shows examples of labeled ground photos as well as the distribution of labeled crop residue fraction data. In corn and soybean fields, conventional tillage has induced significant soil disturbance and limited residue fraction in the field. Meanwhile, for reduced and non-tillage, the fields have much more residue cover left. Fig. 5 (b) is the distribution of crop residue fractions derived from all field photos.

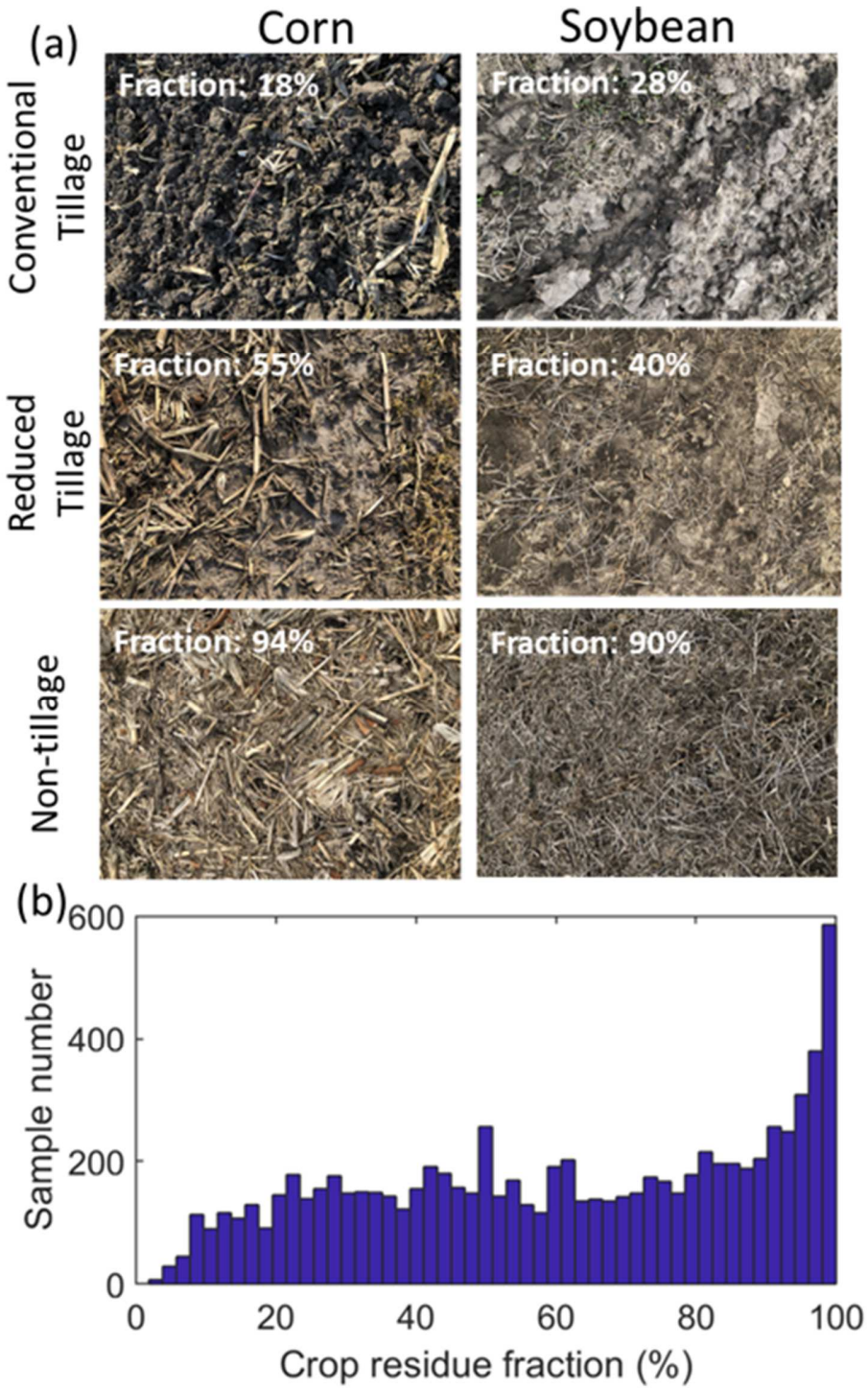


415

416 **Fig. 4** Model testing performance of ResNet-50 to predict crop residue fractions. ResNet-50 provided the

417 first guess of crop residue fractions and then estimates were fine-tuned manually.

418



419

420 **Fig. 5**. Field collected photos and labeled crop residue fractions. (a) Examples of field photos and residue

421 fractions in conventional, reduced, and non-tillage. (b) Histogram of crop residue fraction derived from

422 field photos through image segmentation and labeling.

4.2 Crop residue fraction derived from ground-airborne upscaling

With the paired ground measurements and airborne hyperspectral reflectance in the same location, we developed the CNN model to predict crop residue fractions. The developed CNN model did not show any training overfitting issues (Supplementary Fig. S4). The CNN model shows high predictive performance (Fig. 7a) with $R^2 = 0.82$ and indicates that the hyperspectral-based estimates of crop residue fractions are highly consistent with the ground truth. This accuracy is similar to the previous studies using airborne hyperspectral data to quantify tillage intensity categories (Daughtry et al., 2006). However, this study collected massive ground measurements (6,719 photos) from diverse field conditions and still achieves high performance. The rich ground labeling data enables us to develop robust deep learning models (e.g., CNN model) and to fully utilize the large-volume hyperspectral data in the upscaling process. It is worth mentioning that we have tested different buffer zone sizes for pairing airborne hyperspectral reflectance with ground crop residue measurements, we found that the best performance is achieved with the buffer zone sizes of 3.5m, which may correspond to the GNSS accuracy of field photos. Furthermore, by using the permutation method, we analyzed the feature importance of the CNN model (Fig. 7b). Among all spectral wavelengths, the shortwave infrared wavelengths of 2000-2200 nm have the highest contribution, which agrees with studies using indices such as CAI (Cellulose Absorption Index, Nagler et al., 2003) to quantify crop residue cover. Closely following those wavelengths, the near-infrared wavelengths in 900–1100 nm have high importance. After that, the red and red-edge wavelengths in 600–750 nm have relatively high contributions in predicting crop residue cover. The high feature importance of the shortwave infrared, near-infrared, and red spectral regions agrees with the recent study on using these three wavelengths to calculate crop residue angle index (BAI, Yue et al., 2020) to quantify crop residue cover.

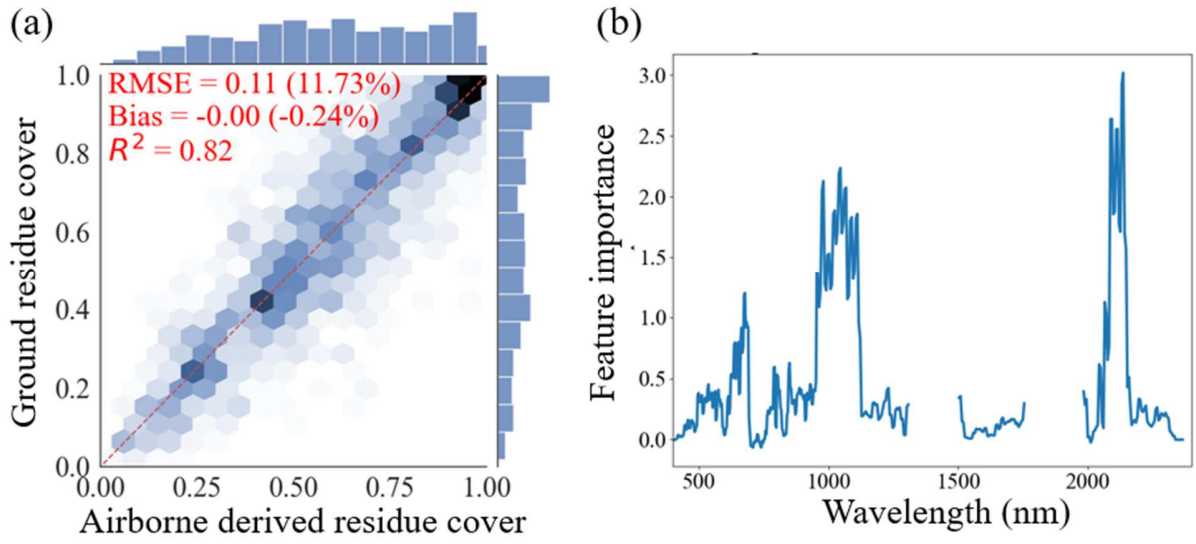


Fig. 6 Upscaling ground-derived crop residue fraction to airborne hyperspectral imagery using Convolutional Neural Networks. (a) Model testing performance, (b) Permutation-based feature importance for each wavelength.

We further compared commonly used airborne data-based spectral indices NDTI and CAI (Fig. 7) with the CNN deep learning results (Fig. 6). Results show that CAI (R^2 = 0.67) performs better than NDTI (R^2 = 0.37), but all these spectral indices perform worse than the deep learning approach (R^2 = 0.82), which utilizes all spectral information in hyperspectral data. The high performance of CAI agrees with the feature importance analysis that 2000-2200 nm have the largest contribution to predicting crop residue cover. The comparison of spectral indices with the convolutional neural network based model performance provides us insights into distinguishing contributions from high spectral and high spatial resolutions in high-performance airborne hyperspectral imagery for quantifying crop residue cover. The performance improvement from CAI to CNN demonstrates that both high spatial resolution and rich high spectral signatures play important roles to highly accurately quantify crop residue cover from airborne hyperspectral data.

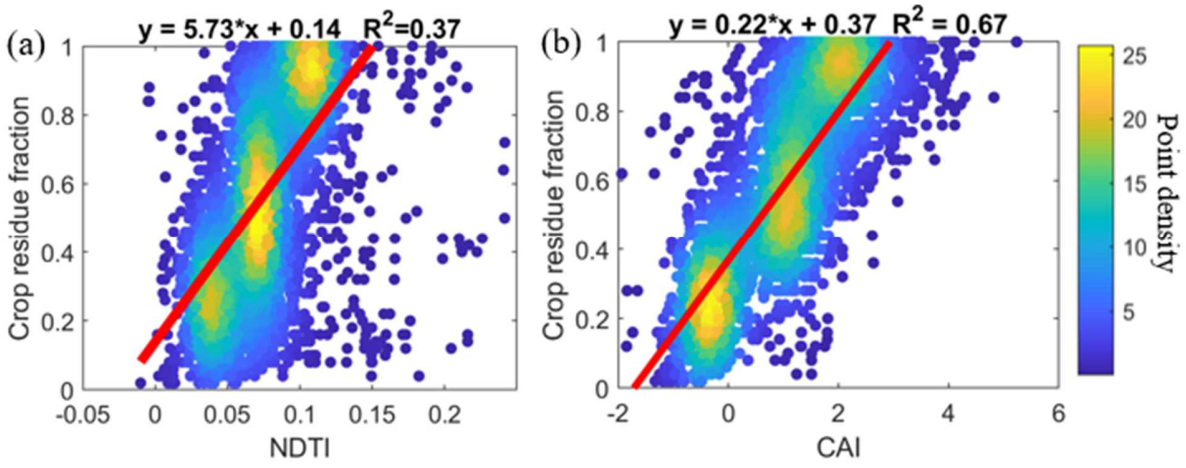


Fig. 7 Comparison of airborne multiple spectral indices to quantify crop residue cover. (a) Normalized Difference Tillage Index (NDTI) and (b) Cellulose Absorption Index (CAI). The linear regression lines are in red. The regression equations and the determination of coefficients (R^2) are in the title of each subplot.

By applying the CNN model to airborne hyperspectral pixels, we quantified crop residue fraction for airborne flight paths (Fig. 8). From the comparison of three fields in Fig. 8, we found that fields A, B, and C correspond to non-tillage, reduced, and conventional tillage with mean crop residue fractions of 0.81, 0.56, and 0.27, respectively.

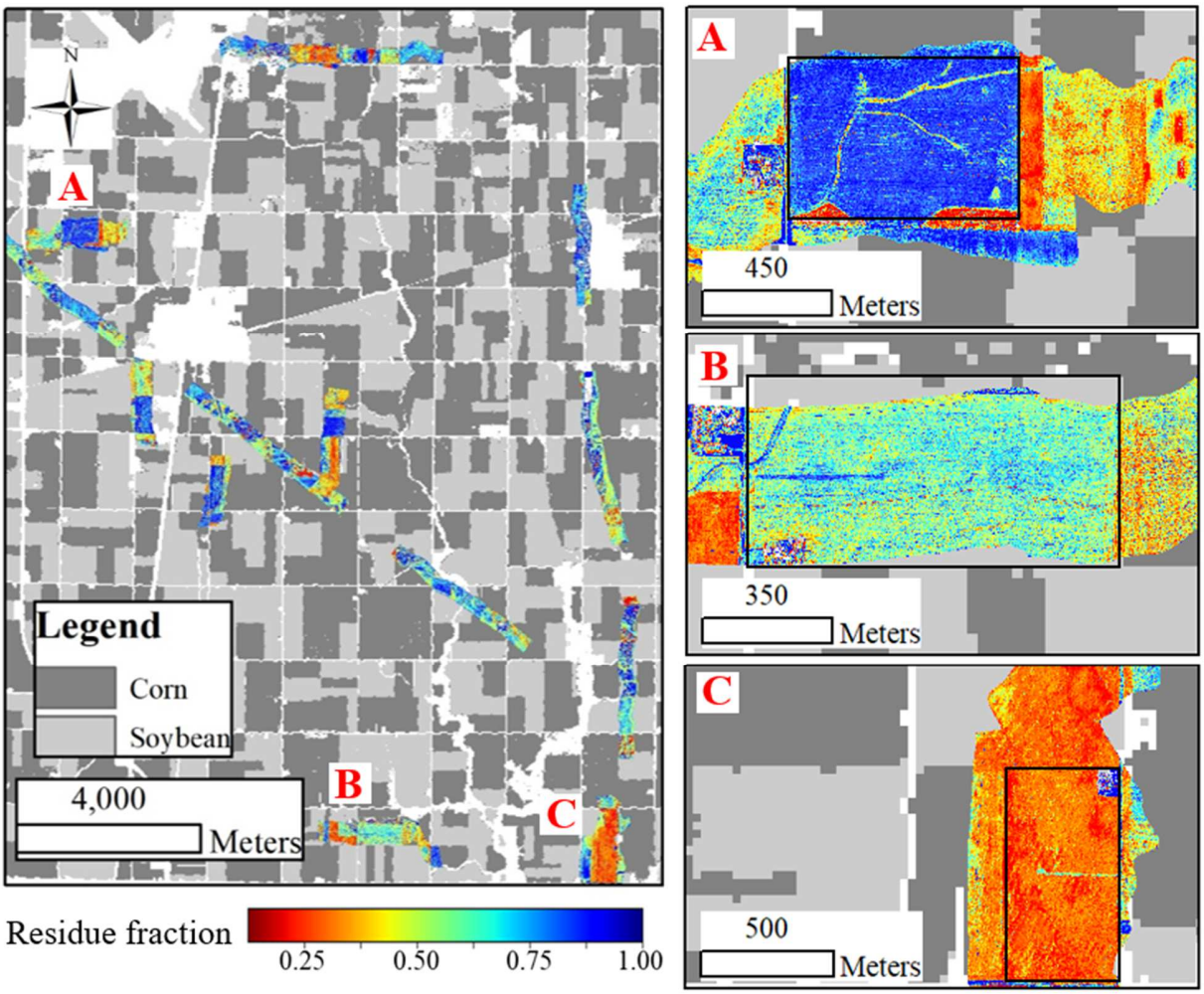


Fig. 8 Examples of quantified crop residue fractions from airborne hyperspectral imagery across Champaign county. The red color refers to the low residue fractions and the blue color indicates the high residue fractions. The base map colors refer to the corn and soybean types. Fields A, B, and C (black rectangles) correspond to non, reduced, and conventional tillage, respectively.

4.3 Crop residue fraction derived from airborne-satellite upscaling

With the airborne acquired 102 km^2 highly accurate quasi "ground truth" of crop residue fraction, we developed the MLP model to integrate airborne estimates with HLS S30 (Sentinel-2). The developed MLP model did not show any overfitting issues (Supplementary Fig. S5). Fig. 9(a) shows the scatterplots of

testing performance between airborne estimates and satellite predictions. The MLP model achieved acceptable performance with $R^2 = 0.73$ and relative RMSE = 13.21%. Such performance is lower than the hyperspectral predictions (Fig. 7) as spectral wavelengths have been largely reduced. Meanwhile, with HLS L30 spectral wavelengths, we found that the model performance (Fig. 9c) is similar to the performance obtained with HLS S30. This shows that the red edge and NIR broadband wavelengths have limited contributions. Furthermore, we utilized the permutation method to analyze the feature importance for the MLP model with HLS S30 and L30 (Fig. 9b and d). The feature importance shows that the highest contributions from these two models are from SWIR 1 and SWIR 2, which agrees with previous studies utilizing NDTI to quantify tillage practices (Daughtry et al., 2006). NIR Narrow and Broad have the third and fourth highest contributions, but MLP models for S30 and L30 achieved similar performance. This shows that NIR Broad may have similar effects as NIR Narrow. Overall, the feature importance analysis for satellite data is generally consistent with the airborne hyperspectral analysis, except for SWIR 1. This indicates that if broadband multispectral data have no sufficient spectral wavelengths in SWIR 2 like airborne hyperspectral data or Worldview-3, the combination of SWIR 1 and 2 is needed to quantify crop residue cover.

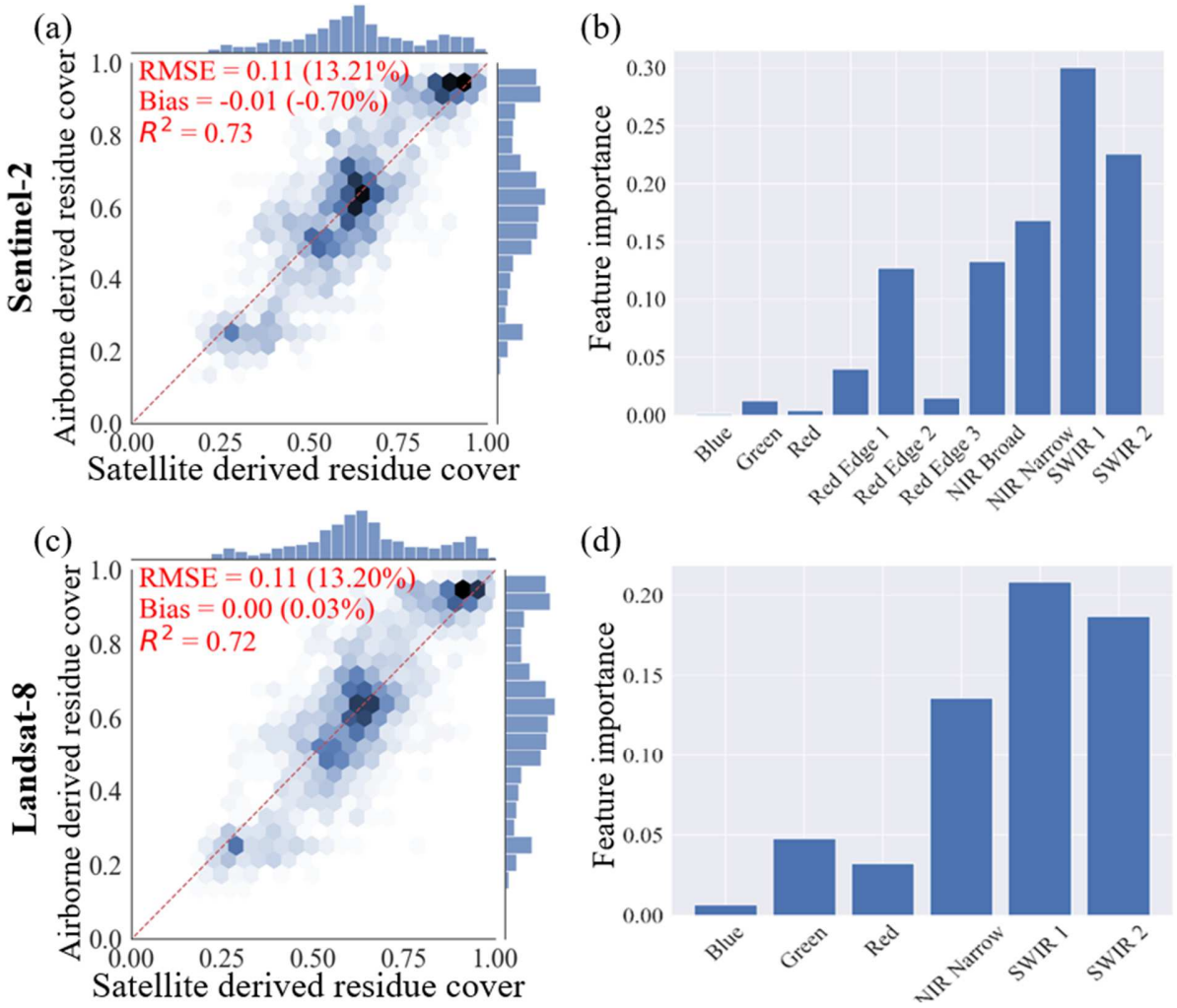
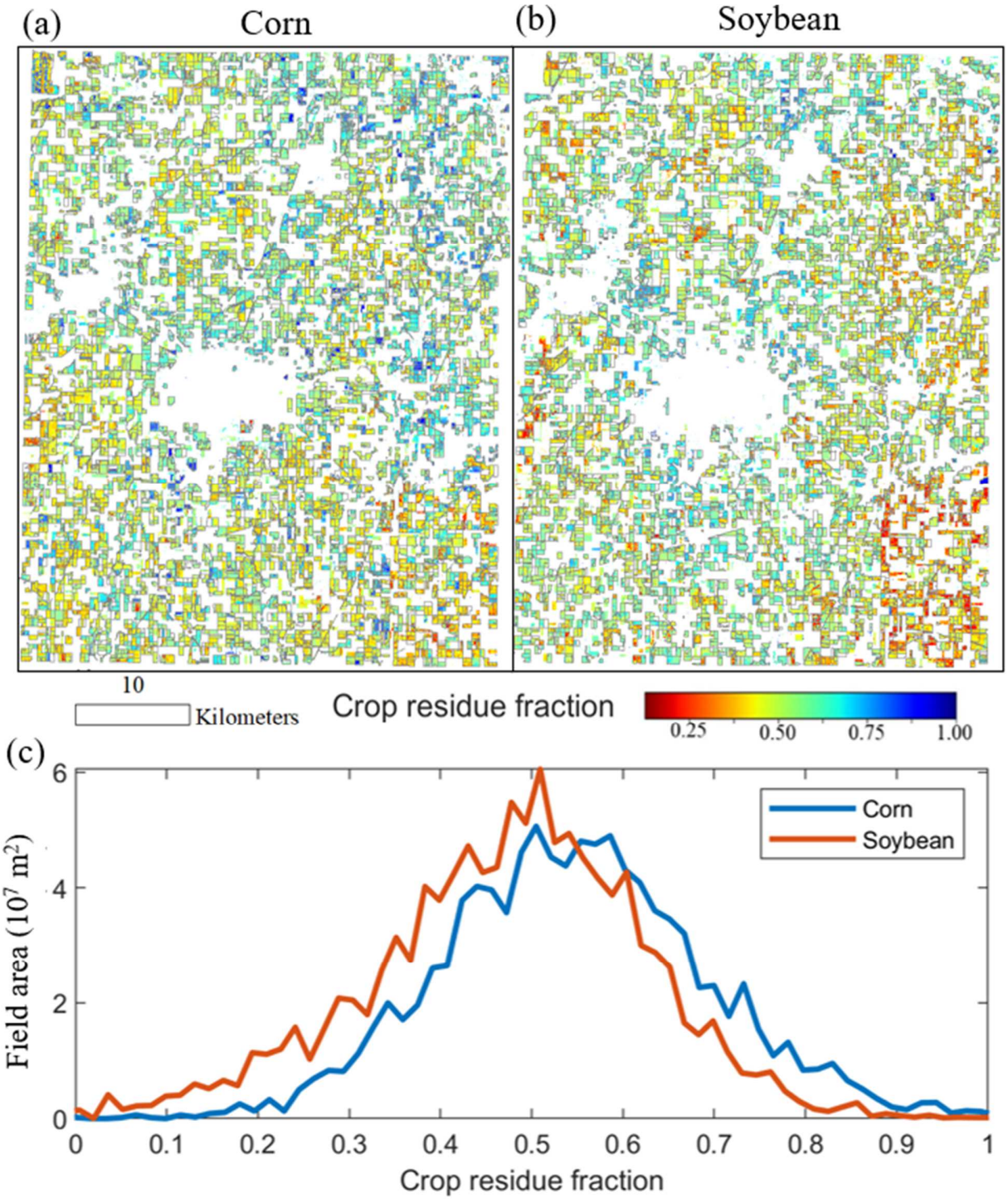


Fig. 9 Upscaling airborne hyperspectral data derived crop residue fraction to satellite HLS multispectral remote sensing using MLP. The first row is the testing performance with HLS Sentinel-2 wavelengths. The second row is the testing performance with HLS Landsat-8 spectral wavelengths. (a) Model testing performance for Sentinel-2 bands, (b) Permutation-based feature importance for Sentinel-2 spectral wavelengths. (c) Model testing performance for Landsat-8 bands, (d) Permutation-based feature importance for Landsat-8 spectral wavelengths.

By applying the MLP model to HLS S30, we obtained the spatial maps for all the corn and soybean fields in Champaign County (Fig. 10). Corn fields overall have relatively higher crop residue fractions than

soybean fields, which may result from the fact that corn has larger biomass and thus larger residues (supplementary Fig. S2).



514 **Fig. 10** Quantified crop residue fraction conditions for every corn and soybean field in Champaign
515 County on December 23th, 2020 from HLS satellite imagery. (a) crop residue fraction for corn fields. (b)
516 crop residue fraction for soybean fields. (c) histogram distribution for crop residue fraction in corn and
517 soybean fields. The field boundaries are derived internally.
518

519 **4.4 Comparison of ground-airborne-satellite and ground-satellite upscaling**

520 To evaluate the benefits of incorporating airborne estimates as an intermediate step, we used the
521 independent ground measurements to test the satellite estimates from the ground-satellite upscaling and
522 ground-airborne-satellite upscaling (Fig. 11). From the scatterplots, the ground-airborne-satellite upscaling
523 scheme achieved much higher performance than the scheme with direct upscaling ground measurements to
524 satellites. The ground-satellite upscaling had good predictions in the high crop residue conditions (Fig.
525 11a), however, failed in predictions in low crop residue conditions. This could be due to the fact that high
526 crop residue fields have similar spectral signatures, while low crop residue fields have diverse soil
527 backgrounds, which is hard to be predicted from multispectral data. Table 2 further summarized the model
528 performance regarding each step upscaling in Fig. 2 and confirms the benefits of airborne hyperspectral
529 imaging as the intermediate step for upscaling ground residue fractions to coarse resolution multispectral
530 imagery.
531

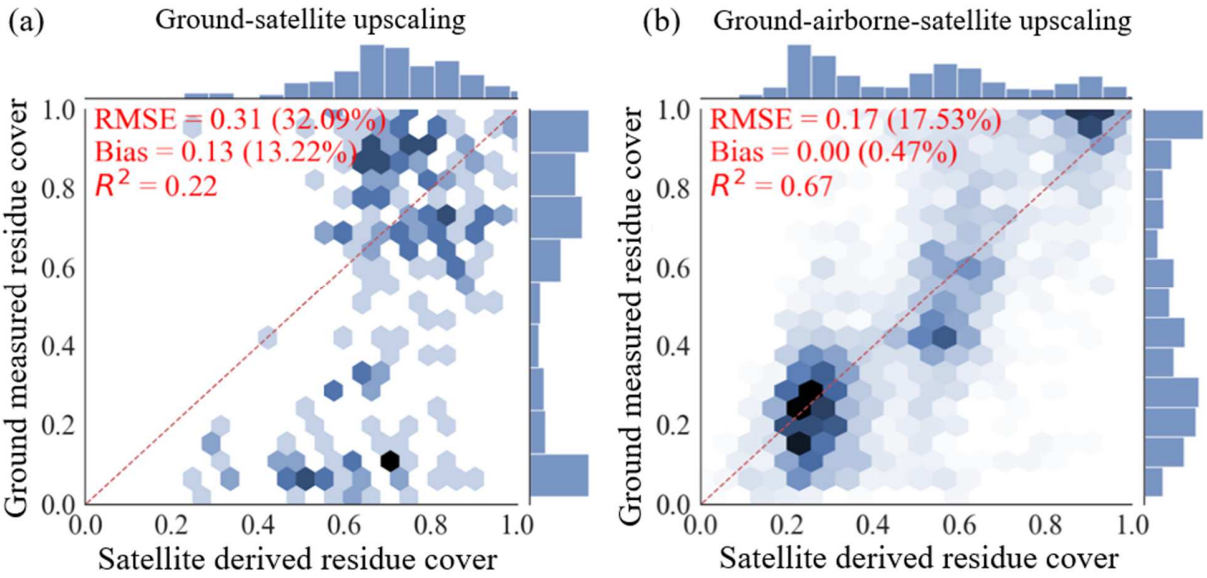


Fig. 11 Testing performance for (a) directly upscaling the ground crop residue fraction to satellite data (b) upscaling through ground-airborne-satellite data.

Table 2. Statistics for comparing ground-airborne-satellite and ground-satellite upscaling methods for quantifying tillage intensity. Ground to airborne refers to airborne model testing performance evaluated by ground data. Airborne to satellite refers to satellite model testing performance evaluated by airborne data. Testing performance refers to satellite model testing performance evaluated by ground data.

Testing performance for upscaling schemes		**R^2**	**RMSE**	**Bias**
Ground-airborne-satellite	Ground to airborne testing	0.82	0.11 (11.73%)	0.00 (-0.24%)
	Airborne to satellite testing	0.73	0.11 (13.21%)	-0.01 (-0.70%)
	Testing performance using ground data	0.67	0.17 (17.53%)	0.00 (0.47%)
Ground-satellite	**Testing performance using ground data**	0.22	0.31 (32.09%)	0.13 (13.22%)

5. Discussion

In this study, we developed a cross-scale sensing framework to integrate ground field photos, airborne hyperspectral imaging, and HLS satellite data through deep learning to quantify regional crop residue fractions. Compared to direct ground-satellite upscaling, the proposed ground-airborne-satellite upscaling framework shows much higher accuracy to quantify crop residue fractions. Moreover, we also found that the shortwave infrared wavelengths in both hyperspectral and multispectral remote sensing data show high contributions to differentiating crop residues and soil. In further discussion, we analyzed the potential improvement in ground photo collection and labeling. The benefits of airborne hyperspectral imaging as an intermediate step for upscaling agriculture ground truth to satellite imagery were also elaborated. We also discussed the potential of future satellite Earth Observation missions for tracking regional tillage practices.

5.1 Improving ground photo collection and labeling

This study spent significant effort to collect a large number of ground photos (6,719) in Champaign and nearby counties to support ground-airborne-satellite upscaling. Compared to highly efficient aircraft and satellite sensing, in-person field investigations could be one key bottleneck for applying the current ground-airborne-satellite framework to broad regions. Other ground data collection approaches such as vehicle-based roadside image data collection (Pilger et al., 2020) or lightweight small unmanned aerial vehicle-based close-range photo collection (Yue and Tian, 2020; Acharya et al., 2021) could potentially further improve ground data collection efficiency. With the well-represented and sufficient ground truth data, we may even not need airborne hyperspectral data as an intermediate step to bridge ground truth and satellite data. Furthermore, this study utilized the modified ResNet-50 regressor and superpixel segmentation to provide the first guess of residue cover and soil/residue masks in the semi-automatic photo labeling pipeline. Nonetheless, this process still requires significant manual checking to ensure data quality. In a few cases, for example, if crop residues are in shadows or soils are too bright, the first guess from ResNet-50 and superpixel segmentation could underestimate or overestimate crop residue cover, respectively. With

the acquisition of more and more ground tillage photos, deep-learning-based supervised image segmentation could be potentially applied to better distinguish soil and crop residue pixels and obtain more accurate residue labels. In addition, currently all ground photos were manually and carefully checked. Multiple group label comparison experiments could be helpful to understand and possibly improve the accuracy of label data.

5.2 Benefits of airborne hyperspectral data as an intermediate step for upscaling

With the rapid development of space technology, satellites offer petabyte-scale diverse Earth Observation data to monitor agricultural ecosystem dynamics (Guan et al., 2017). Meanwhile, the recent progress in deep learning algorithms also empowers our ability to interpret these petabyte-scale satellite data (Reichstein et al., 2019). However, deep learning models with a large volume of satellite data as inputs require sufficient and massive label data for model training and testing. Moreover, as the intrinsic tradeoffs among spectral, spatial, temporal, and radiometric resolutions for satellite data (Ustin and Middleton, 2021), satellite missions such as Landsat and Sentinel-2 with 10–30 m spatial resolution and 6–10 bands are popularly used for agricultural monitoring. Nonetheless, with such spatial resolutions, field measurements from small sampling units (often < 5 m^2) usually have significant uncertainties to be matched with satellite pixels. Therefore, lacking high-quality, large-volume, high-resolution, and cost-effective ground truth data to support satellite model development becomes the main bottleneck. To address this challenge, airborne hyperspectral imaging systems offer three major benefits to support the upscaling from ground measurements to satellite prediction. First, airborne hyperspectral data offer hundreds of spectral wavelengths, which are closely linked with surface biophysical and biochemical parameters, and are able to be integrated with representative ground data through deep learning to obtain highly accurate agroecosystem variables. These quantified agroecosystem variables inferred from airborne hyperspectral systems have sufficient accuracy to be treated as quasi "ground truth". For example, this study demonstrated that crop residue fractions and tillage intensity were highly accurately predicted by airborne hyperspectral data with CNN (R^2 = 0.82, relative RMSE = 11.73%). Second, airborne data can build the bridge to address

591 the scaling-mismatch issues. Airborne data have a fine spatial resolution to be seamlessly matched with
592 ground data down to sub-meter level, and airborne quasi-ground truth can be further resampled to the same
593 resolution as satellite pixels (30 m in this study) to avoid ground-satellite data mismatch in the conventional
594 approaches. The direct upscaling ground measurements to satellite data showed weak performance (Fig.
595 11a) in this study, partly due to the insufficient ground photos (6–8 photos per satellite pixel) to represent
596 the whole satellite pixel condition. High spatial resolution airborne data can be very useful to support
597 ground sampling to obtain spatially detailed intermediate data for upscaling to coarse-resolution satellite
598 data in a cost-effective manner. Third, with flight permission and suitable weather conditions, airborne
599 hyperspectral systems could be flexibly deployed to collect surface reflectance across a large region to
600 generate large-volume label data to support large-scale satellite upscaling. Airborne hyperspectral imaging
601 is a powerful tool for agricultural ground-truthing to augment ground measurements in a high-throughput
602 manner. As such, airborne highly accurate estimates are able to be further upscaled with satellite
603 observations and deep learning to derive agroecosystem variables for every field in a given region, as an
604 example demonstrated in this study on crop residue cover monitoring.
605

606 **5.3 Satellite data for monitoring crop residue cover**

607 This study demonstrated the utility of HLS data to detect regional crop residue fractions. From feature
608 importance analysis, we found that shortwave infrared bands are vital for detecting crop residue cover.
609 However, we also found clear performance degradation from airborne hyperspectral data to satellite
610 multispectral data (Table 2). This agrees with previous studies that hyperspectral remote sensing
611 outperforms multispectral sensing to detect crop residue fraction, and narrowband shortwave infrared
612 wavelengths are needed for crop residue detection (Daughtry, et al. 2008; Hively et al., 2018). The current
613 Landsat and Sentinel-2 only have two wavelengths in the shortwave infrared range and could be a limiting
614 factor for further improving satellite detection accuracy. With the new and forthcoming satellite
615 hyperspectral missions or multispectral missions with rich shortwave infrared wavelengths, these missions

such as PRISMA (Stefano et al., 2013), EnMAP (Guanter et al., 2009), NASA SBG (Cawse-Nicholson et al., 2021) and Landsat Next (Hively et al., 2021) can provide great capabilities to detect crop residues globally.

For the current Landsat or Sentinel-2 multispectral data, uncertainties in soil background signals from variable soil properties, moisture conditions, and crop residue moisture are major challenges to accurately detecting the fractions of crop residue cover (Serbin et al., 2009; Yue et al., 2020). For example, the direct ground-satellite upscaling cannot predict low crop residue conditions well, as diverse soil signals across fields (Fig. 11a). To conquer these challenges, this study conducted airborne hyperspectral surveys to scan 102 km^2 of croplands across this region. These collected massive information on crop residue and soil fractions supported satellite multispectral data to distinguish crop residues and soils across diverse field conditions. In addition, incorporating soil property or other remote sensing data could further help to improve the predictions of crop residue cover. For example, gSSURGO soil property database (soil color proxy) and Sentinel-1 C-band Synthetic Aperture Radar backscattering (surface roughness or soil moisture proxy) could be added to the satellite modeling framework. Furthermore, satellite multi-temporal techniques to utilize time-series observations to minimize soil moisture effects and exclude soil background signals are also promising to improve the current satellite detection. For example, Zheng et al. (2012) found that the temporal minimum NDTI approach is effective to exclude soil moisture effects to quantify crop residue cover. In addition, the recently proposed novel indices such as Nonphotosynthetic Vegetation-Soil Separation Index (NSSI, Tian et al., 2021) and Broadband crop residue Angle Index (BAI, Yue et al., 2020) also show capability to reduce soil moisture effects to quantify crop residue cover.

5.4 Scalability of ground-airborne-satellite integrative sensing

Accurate information on crop residue cover is valuable to guide conservation practices as well as understanding nutrient dynamics in agroecosystems. The scalable and cost-effective approach to quantify crop residue cover from remote sensing is urgently needed. Compared to the direct ground-satellite

upscaling approach, ground-airborne-satellite upscaling added one more step to airborne hyperspectral sensing, but achieved much higher accuracy. Though adding this one more step, from the scalability perspective, deploying an airborne hyperspectral imaging system to collect massive quasi-ground truth of residue cover could be still much more cost-effective and scalable than conventional ground sampling. For example, this study conducted three airborne flights to acquire 102 km^2 cropland hyperspectral reflectance to obtain residue cover information, which is much more high-throughput than in-person field investigations. Furthermore, as incorporating diverse field conditions, the spectral relationship trained from ground residue cover to airborne data is generic, and would change little over space and time in this region. For example, to deploy such approaches to broader space and time domains with similar field conditions, we do not expect that more airborne surveys are needed for different years. If the soil types are different in the new geography, additional strategic airborne sampling could be needed for fine-tuning remote sensing models to improve the accuracy and generalizability of the algorithms. Overall, the ground-airborne-satellite algorithm is highly scalable. The developed cross-scale sensing framework and methodology, though only trained in the study region (central Illinois), should have broad transferability and applicability to large corn and soybean cropland geographies with fine-tuning through only a small amount of extra ground truth, making this method highly scalable and cost-effective. Ultimately, we can only use satellite data to scale up to broader regions with their local ground data for spectral model fine-tuning.

Beyond crop residues, timely data on non-photosynthetic vegetation in agroecosystems indicates crop stress, environmental conditions, and timing for harvest. In natural ecosystems, high-resolution information on non-photosynthetic components is also vital for the diagnosis of ecosystem health, fire risks, and biogeochemical processes. Given the importance, non-photosynthetic vegetation has been identified as a priority variable in the context of new spaceborne imaging spectroscopy missions (Berger et al., 2021). To deliver operational products for non-photosynthetic vegetation at the global scale, besides airborne-satellite integrative sensing and advanced data-driven machine learning models in this study, we also need to further

develop soil-vegetation radiative transfer models (Berger et al., 2021) or spectral indices (e.g., Tian et al., 2021) to enable the direct retrieval of non-photosynthetic vegetation from multi-scale remote sensing.

6. Conclusion

Accurately quantifying crop residue fraction and tillage intensity is vital for evaluating public conservation programs as well as understanding agroecosystem biogeochemical processes. However, traditional remote sensing approaches are challenging to obtain regional-scale detailed information on crop residue fractions. Given the urgent needs, this study developed a novel cross-sensing framework to integrate proximal sensing, airborne hyperspectral imaging, and satellite multispectral sensing with the aid of deep learning to quantify crop residue fractions in the central region of the U.S. Corn Belt. Using image segmentation, we labeled 6,719 ground photos from fields across the study region. By matching these ground measurements with airborne hyperspectral imagery, we utilized CNN to accurately quantify crop residue fraction (R^2 = 0.82, relative RMSE = 11.73%) to effectively augment the quasi "ground truth" to support satellite upscaling to quantify all corn and soybean tillage intensity. Compared to the direct ground-satellite upscaling framework (R^2 = 0.35, relative RMSE = 21.91%), this ground-airborne-satellite integrative sensing framework achieved much better performance to quantify crop residue fractions (R^2 = 0.67, relative RMSE = 17.53%). Among all wavelengths, we found that shortwave infrared 2000-2200 nm has a high contribution to crop residue predictions. This study demonstrated the effectiveness of airborne hyperspectral imaging as agricultural "ground truth" collectors to fill the scale-mismatch between petabyte-scale satellite Earth Observation and plot-level ground measurements to seamlessly integrate multi-source data to advance the monitoring of sustainable agriculture.

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Cross-scale sensing for detecting tillage intensity

- Airborne hyperspectral imaging as an intermediate step
- High scalability: Closing gaps from ground measurements to satellite observations
- Large-scale detection
- Cost-effective

Region

Every field

Tier 3: Satellite

Landscape

Ten thousands hectare

Tier 2: Airborne

Point

Hundreds of fields

Tier 1: Ground photos

